Capstone Project Summary: Customer Churn Prediction

1. Business Problem: Telecom companies face immense revenue loss due to customer churn. It is 5x more expensive to acquire a new customer than to retain an existing one. Our goal is to predict churn accurately.

2. Workflow: We engineered meaningful business features (Tenure Groups, Spending Categories, Risk Indicators). We handled data preprocessing strictly inside Scikit-Learn Pipelines to prevent data leakage.

3. Results: XGBoost/Random Forest emerged as top predictors. Through Hyperparameter tuning via GridSearchCV, we maximized ROC-AUC to ensure robust identification of churn candidates.

4. Explainable AI: Using SHAP values, we decoded the Blackbox model. Month-to-month contracts, short tenure, and lack of technical support are the primary catalysts for churn.

5. Recommendations: The business should immediately target the Month-to-Month segment with loyalty lock-in incentives and deploy the predict.py script in a daily batch pipeline to flag at-risk accounts.